

Design and Simulation of a Wireless University Campus Network using Cisco Packet Tracer.

A newly established university plans to implement a **campus-wide wireless network infrastructure** to support academic, administrative, and hostel services. The university contains several buildings including academic blocks, a library, dome building, IT consulting center, and hostel areas.

The IT department has decided to implement a **centralized network architecture** with routers, switches, servers, and wireless access points to ensure reliable communication and internet access across the campus.

The network must provide the following services:

- Web services for university websites
- Email communication between staff and students
- Domain name resolution using DNS
- Wireless internet connectivity for all buildings
- Secure router access using SSH
- Password-protected wireless networks

The network should be divided into **two (02) main areas**

1. Campus Area

Includes:

- **Academic Block 1 (AB1)** –

2 Laptops and 2 Desktops with an AP
(IP: 192.168.1.14 – 192.168.1.17)

- **Academic Block 2 (AB2)** –

2 Laptops and 2 Desktops with an AP

(IP: 192.168.1.10 – 192.168.1.13)

- **Dome Building**

1 Laptop and 2 Desktops with an AP

(IP: 192.168.1.2 – 192.168.1.4)

- **Library** – 2 Desktops with an AP (IP: 192.168.1.5 – 6)
- **IT Consulting Center** – A Laptop and 2 Desktops with an AP (IP: 192.168.1.7 – 9)
- **Server Center** – 03 Types of servers connected to a switch (IP: 192.168.2.3 – 5)

2. Hostel Area

Includes:

- **Boys Hostel Block** – 2 Laptops, a Desktop and a Smartphone with an AP
- **Girls Hostel Block** – 2 Laptops, a Desktop and a Smartphone with an AP

Each sub area should have its **own wireless access point and IP addressing scheme**.

The network must ensure that **all devices across the campus can communicate with servers and each other**.

Tasks

Students are required to perform the following tasks using **Cisco Packet Tracer**.

Task 1 – Network Topology Design

Design a complete university campus network using the following devices:

The topology must include:

- Main Router (Core Router)
- Campus Router
- Hostel Router
- Server Switch
- Campus Switch
- Hostel Switch

Hint: All 03 Routers should be in a serial connection (**DCE**).

Server Network

Network: 192.168.2.0/24

Configure:

- DNS Server
- Web Server
- Email Server

Hint: All 03 Servers should be connected to the **Server Switch** along with the **Main Router**.

Hostel Network

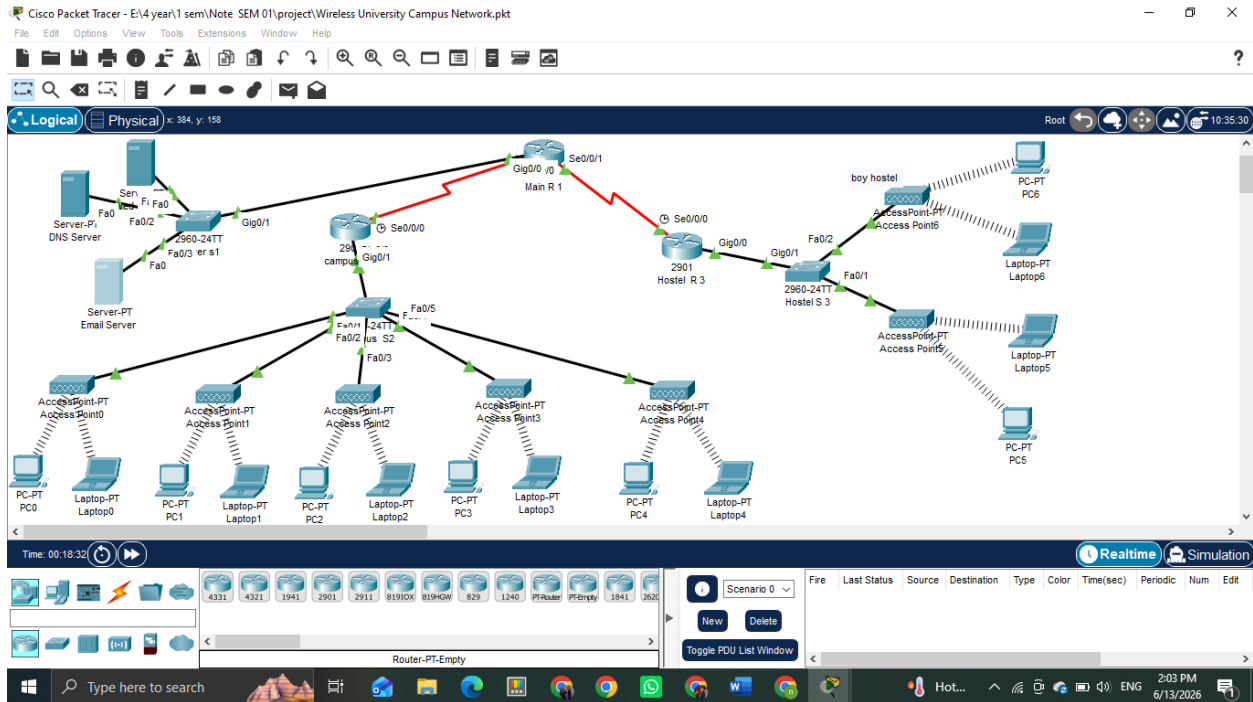
Network: 192.168.3.0/24

Assign IP addresses for:

- Boys Hostel - (IP- 192.168.3.6 – 9)
- Girls Hostel - (IP- 192.168.3.2 – 5)

Default Gateway: 192.168.3.1

Hint- All 02 Hostels should be connected to Hostel Switch along with Hostel Router

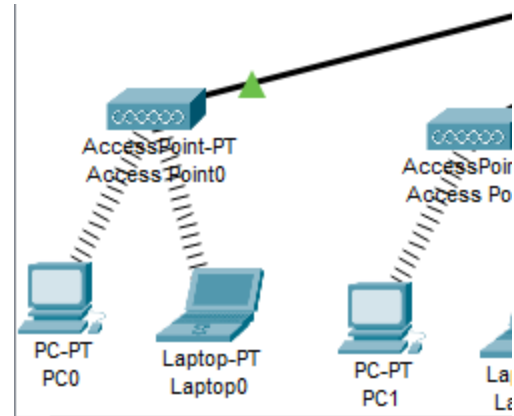
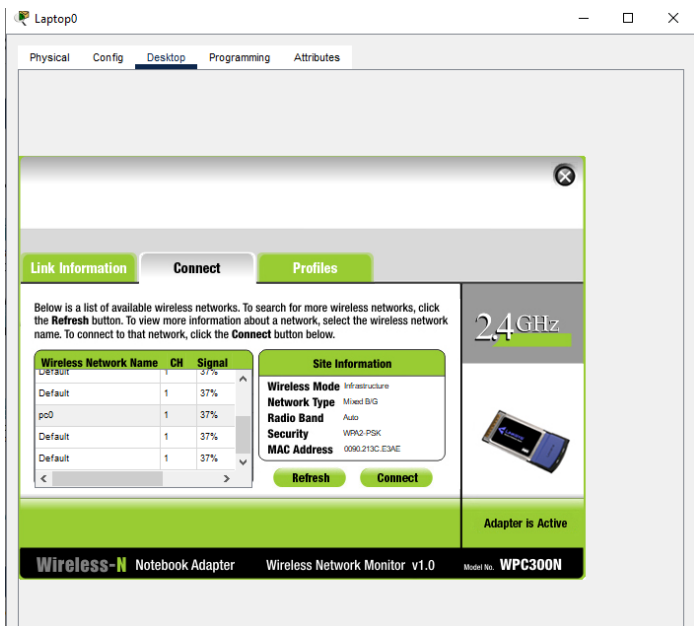
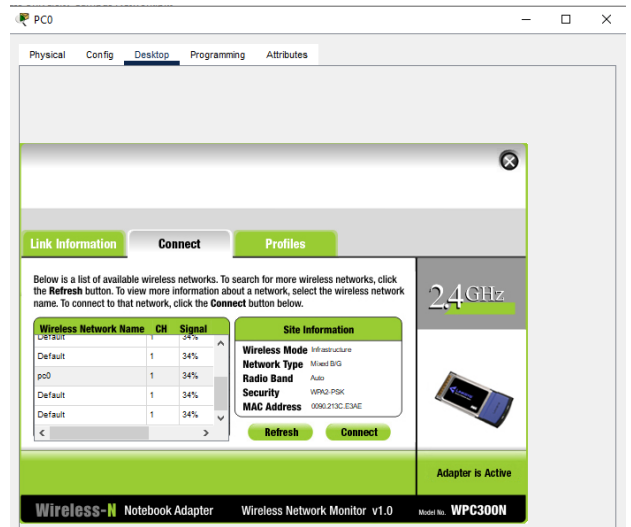
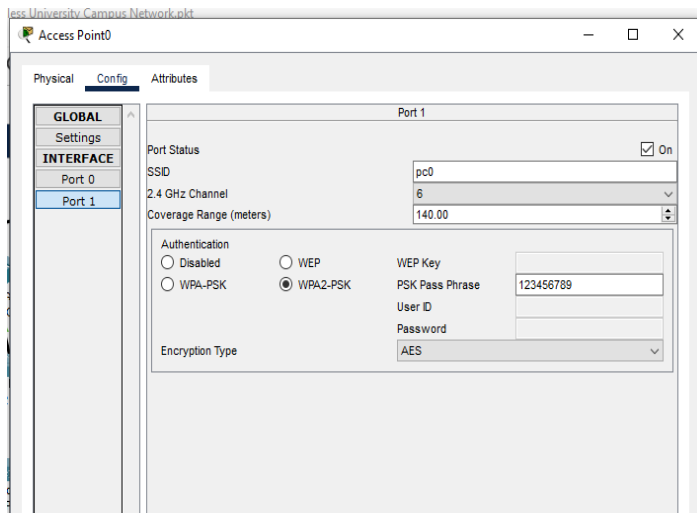


Task 2 – Wireless Network Configuration

Create a table for all Access Points with **Location**, **SSID** and secure them with **passwords** -

(SSID and Passwords should be Unique and standard)

Connect laptops and smartphones to the wireless network.



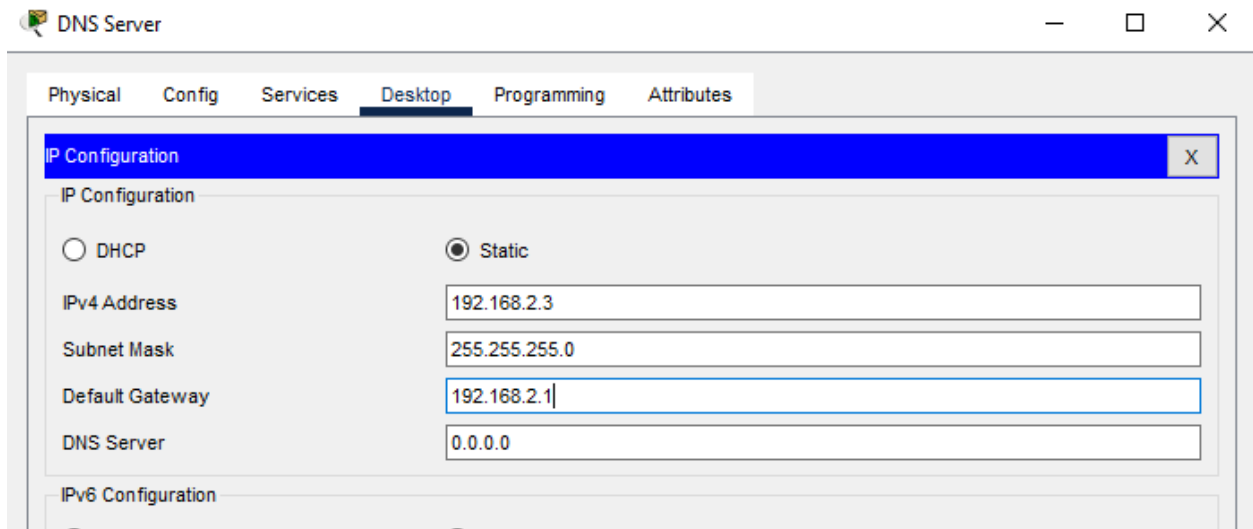
Task 3 – Server Configuration

Configure the following services:

DNS Server

Create domain records such as:

- **www.university.edu** - 192.168.2.4
- **mail.university.edu** - 192.168.2.5
-



The screenshot shows a window titled "DNS Server" with a standard Windows-style title bar (minimize, maximize, close buttons). The window has a tabbed interface with tabs labeled "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Desktop" tab is currently selected. Inside the window, there is a section titled "IP Configuration" with a blue header bar and a close button (X). Below this header, there are two radio buttons: "DHCP" (unselected) and "Static" (selected). Under the "Static" configuration, there are four text input fields: "IPv4 Address" containing "192.168.2.3", "Subnet Mask" containing "255.255.255.0", "Default Gateway" containing "192.168.2.1", and "DNS Server" containing "0.0.0.0". Below these fields, there is a section for "IPv6 Configuration" which is currently collapsed.

DNS Server

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

●

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

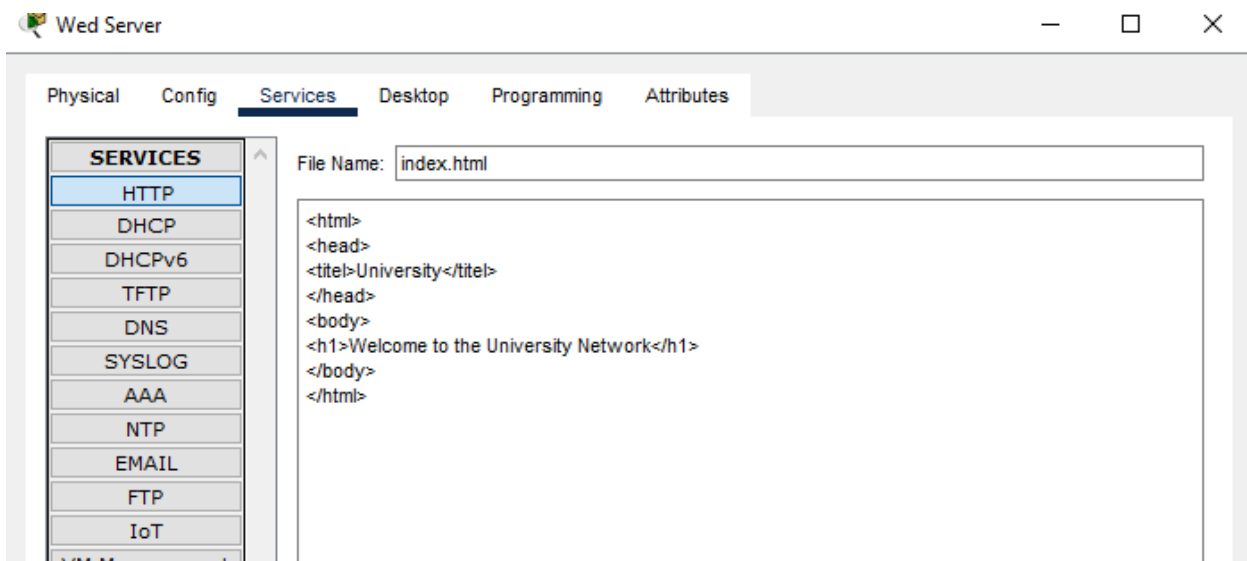
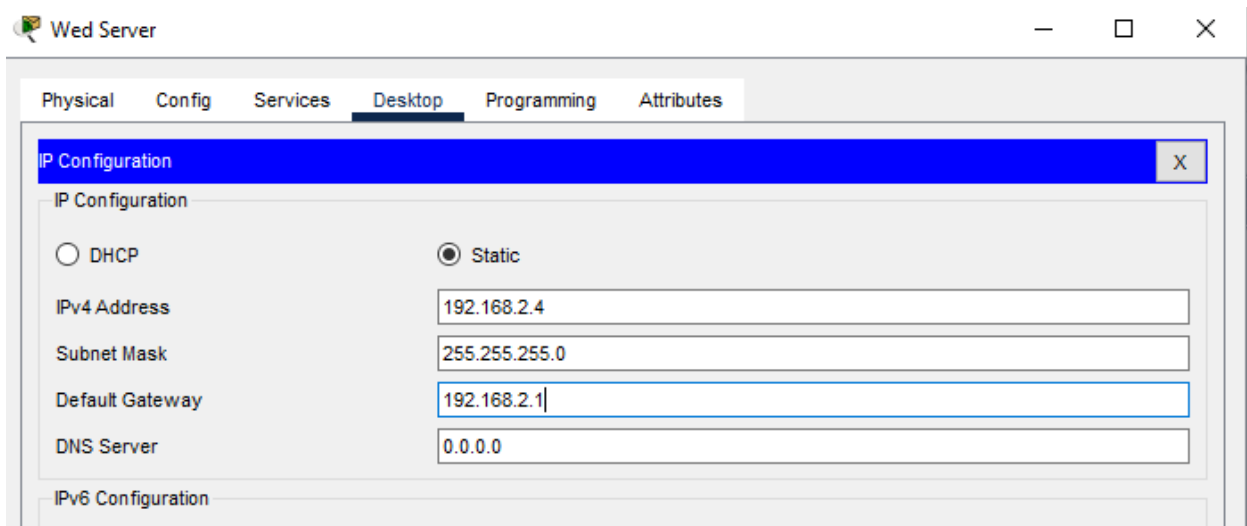
No.	Name	Type	Detail
0	mail.university.edu	A Record	192.168.2.5
1	www.university.edu	A Record	192.168.2.4

Web Server

Create a simple web page displaying:

HEADER - *"Welcome to the University Network"*

CONTENT - image.jpg (Use any sample image or Screen Shot from your pc)



Email Server

Create email accounts such as:

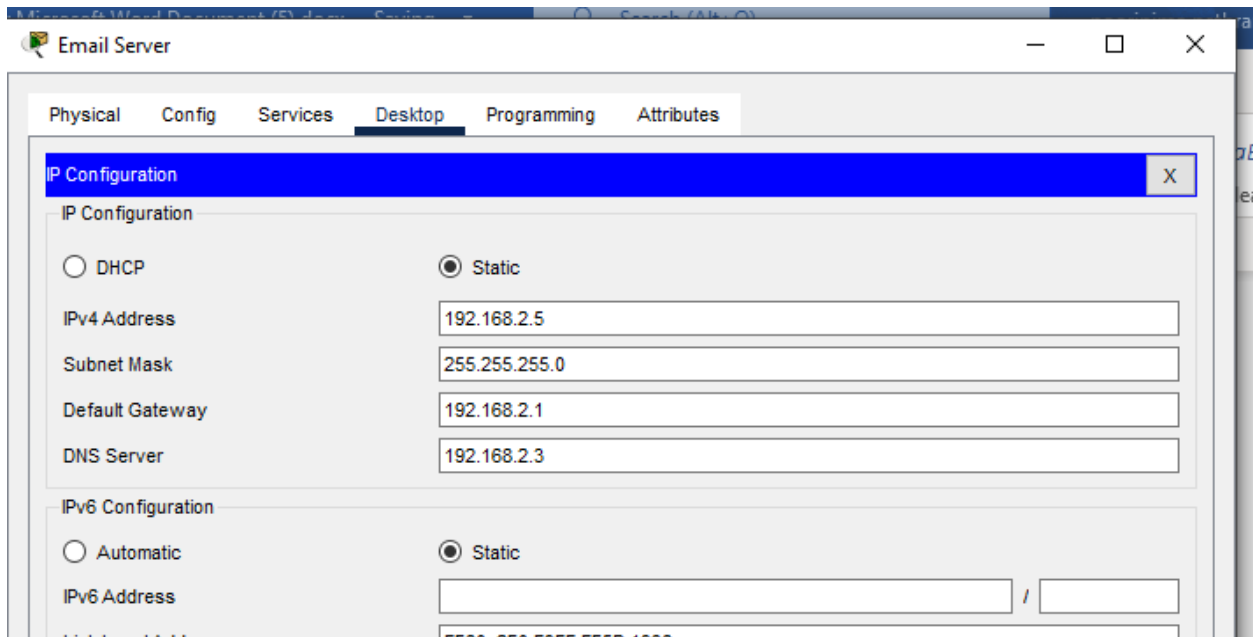
- **student1@university.edu**
- **staff1@university.edu**

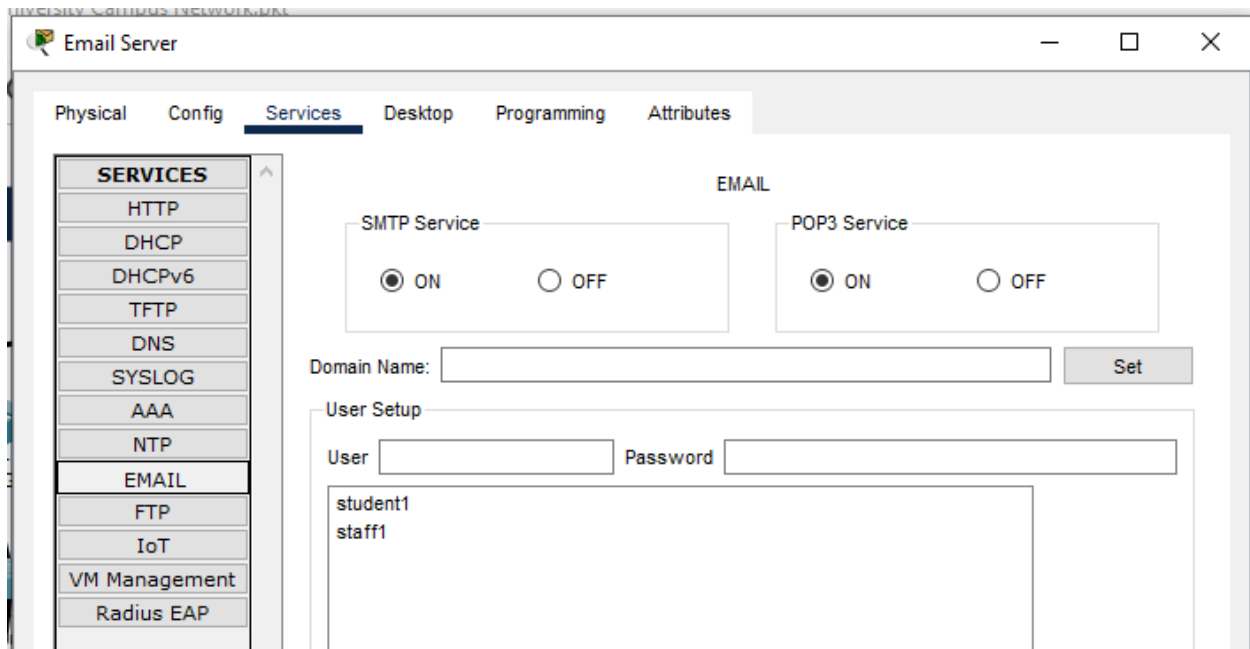
Hint – Refer the below table for user account creation

User	Password
student1	123
staff1	123

After creation of 02 accounts use a **Boys hostel PC** as a **student** and a **AB1 laptop** as a **staff** to show student **send a mail** containing the message of "**I'm SEU/IS/20/ICT/XXX**" to staff (with proof).

In other side staff **received a mail** from student (with proof).





Task 4 – Router Configuration

Configure routers with:

- Hostnames – (refer job 1 for Hostnames)
- Interface IP addresses

Hint - Interface Addressing Information

Device	Interface	IP Address	Subnet Mask	Description
MainRouter	GigabitEthernet0/0	192.168.2.1	255.255.255.0	Server Network
MainRouter	Serial0/0/0	10.0.0.1	255.255.255.252	Link to Campus Router
MainRouter	Serial0/0/1	10.0.0.5	255.255.255.252	Link to Hostel Router
CampusRouter	GigabitEthernet0/0	192.168.1.1	255.255.255.0	Campus LAN
CampusRouter	Serial0/0/0	10.0.0.2	255.255.255.252	Link to Main Router
HostelRouter	GigabitEthernet0/0	192.168.3.1	255.255.255.0	Hostel LAN
HostelRouter	Serial0/0/0	10.0.0.6	255.255.255.252	Link to Main Router

- Static routing between networks

- Console password
- SSH authentication

Hint - Security Configuration

Router	Console Password	SSH Password
Main Router	cisco	admin
Campus Router	uni@123	admin
Hostel Router	uni@123	admin

Main Router

 Main R 1

Physical Config CLI Attributes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname MainRouter
MainRouter(config)#
MainRouter(config)#interface gigabitEthernet0/0
MainRouter(config-if)#ip address 192.168.2.1 255.255.255.0
MainRouter(config-if)#no shutdown

MainRouter(config-if)#
MainRouter(config-if)#interface serial0/0/0
MainRouter(config-if)#ip address 10.0.0.1 255.255.255.252
MainRouter(config-if)#clock rate 64000
This command applies only to DCE interfaces
MainRouter(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
MainRouter(config-if)#
MainRouter(config-if)#interface serial0/0/1
MainRouter(config-if)#ip address 10.0.0.5 255.255.255.252
MainRouter(config-if)#clock rate 64000
This command applies only to DCE interfaces
MainRouter(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
MainRouter(config-if)#
MainRouter(config-if)#ip route 192.168.1.0 255.255.255.0 10.0.0.2
MainRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.6
MainRouter(config)#
MainRouter(config)#line console 0
MainRouter(config-line)#password cisco
MainRouter(config-line)#login
MainRouter(config-line)#
MainRouter(config-line)#ip domain-name university.edu
MainRouter(config)#username admin secret admin
MainRouter(config)#
MainRouter(config)#crypto key generate rsa
```

```
MainRouter(config-line)#login
MainRouter(config-line)#
MainRouter(config-line)#ip domain-name university.edu
MainRouter(config)#username admin secret admin
MainRouter(config)#
MainRouter(config)#crypto key generate rsa
The name for the keys will be: MainRouter.university.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

MainRouter(config)#
*Mar 1 0:7:44.778: %SSH-5-ENABLED: SSH 1.99 has been enabled
MainRouter(config)#line vty 0 4
MainRouter(config-line)#login local
MainRouter(config-line)#transport input ssh
MainRouter(config-line)#
MainRouter(config-line)#end
MainRouter#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
MainRouter#
MainRouter#
MainRouter#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

MainRouter#
```

Campus Router

 campus R 2

Physical Config CLI Attributes

IOS

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname CampusRouter
CampusRouter(config)#
CampusRouter(config)#interface gigabitEthernet0/0
CampusRouter(config-if)#ip address 192.168.1.1 255.255.255.0
CampusRouter(config-if)#no shutdown

CampusRouter(config-if)#
CampusRouter(config-if)#interface serial0/0/0
CampusRouter(config-if)#ip address 10.0.0.2 255.255.255.252
CampusRouter(config-if)#no shutdown

CampusRouter(config-if)#
CampusRouter(config-if)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
CampusRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.1
CampusRouter(config)#
CampusRouter(config)#line console 0
CampusRouter(config-line)#password uni@123
CampusRouter(config-line)#login
CampusRouter(config-line)#
CampusRouter(config-line)#ip domain-name university.edu
CampusRouter(config)#username admin secret admin
CampusRouter(config)#
CampusRouter(config)#crypto key generate rsa
The name for the keys will be: CampusRouter.university.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

```

CampusRouter(config)#line console 0
CampusRouter(config-line)#password uni@123
CampusRouter(config-line)#login
CampusRouter(config-line)#
CampusRouter(config-line)#ip domain-name university.edu
CampusRouter(config)#username admin secret admin
CampusRouter(config)#
CampusRouter(config)#crypto key generate rsa
The name for the keys will be: CampusRouter.university.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

CampusRouter(config)#
*Mar 1 0:6:46.199: %SSH-5-ENABLED: SSH 1.99 has been enabled
CampusRouter(config)#line vty 0 4
CampusRouter(config-line)#login local
CampusRouter(config-line)#transport input ssh
CampusRouter(config-line)#
CampusRouter(config-line)#end
CampusRouter#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
CampusRouter#
CampusRouter#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

```

Hostel Router

 Hostel R3

Physical Config CLI Attributes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname HostelRouter
HostelRouter(config)#
HostelRouter(config)#interface gigabitEthernet0/0
HostelRouter(config-if)#ip address 192.168.3.1 255.255.255.0
HostelRouter(config-if)#no shutdown

HostelRouter(config-if)#
HostelRouter(config-if)#interface serial0/0/0
HostelRouter(config-if)#ip address 10.0.0.6 255.255.255.252
HostelRouter(config-if)#no shutdown

HostelRouter(config-if)#
HostelRouter(config-if)#ip route 192.168.1.0 255.255.255.0 10.0.0.5
HostelRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.5
HostelRouter(config)#
HostelRouter(config)#line console 0
HostelRouter(config-line)#password uni@123
HostelRouter(config-line)#login
HostelRouter(config-line)#
HostelRouter(config-line)#ip domain-name university.edu
HostelRouter(config)#username admin secret admin
HostelRouter(config)#
HostelRouter(config)#crypto key generate rsa
The name for the keys will be: HostelRouter.university.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

HostelRouter(config)#
*Mar 1 0:12:3.310: %SSH-5-ENABLED: SSH 1.99 has been enabled
HostelRouter(config)#line vty 0 4
HostelRouter(config-line)#login local
```



```
HostelRouter(config)#crypto key generate rsa
The name for the keys will be: HostelRouter.university.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

HostelRouter(config)#
*Mar 1 0:12:3.310: %SSH-5-ENABLED: SSH 1.99 has been enabled
HostelRouter(config)#line vty 0 4
HostelRouter(config-line)#login local
HostelRouter(config-line)#transport input ssh
HostelRouter(config-line)#
HostelRouter(config-line)#end
HostelRouter#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
|
```
